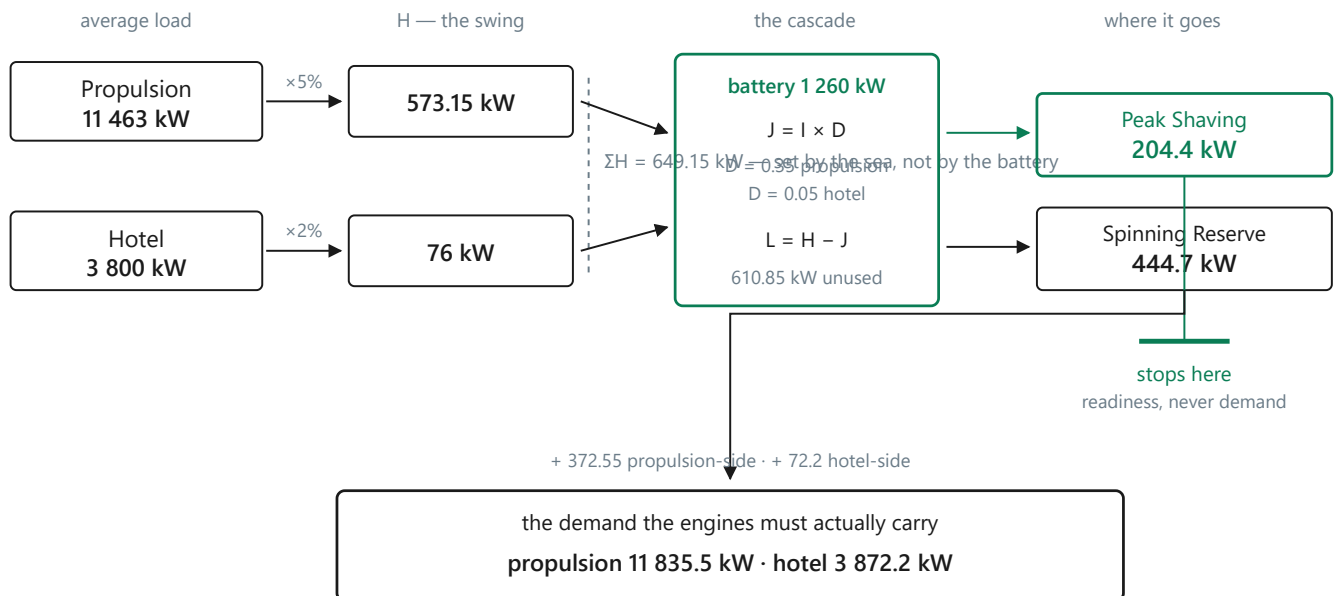


00a — The pipeline, drawn

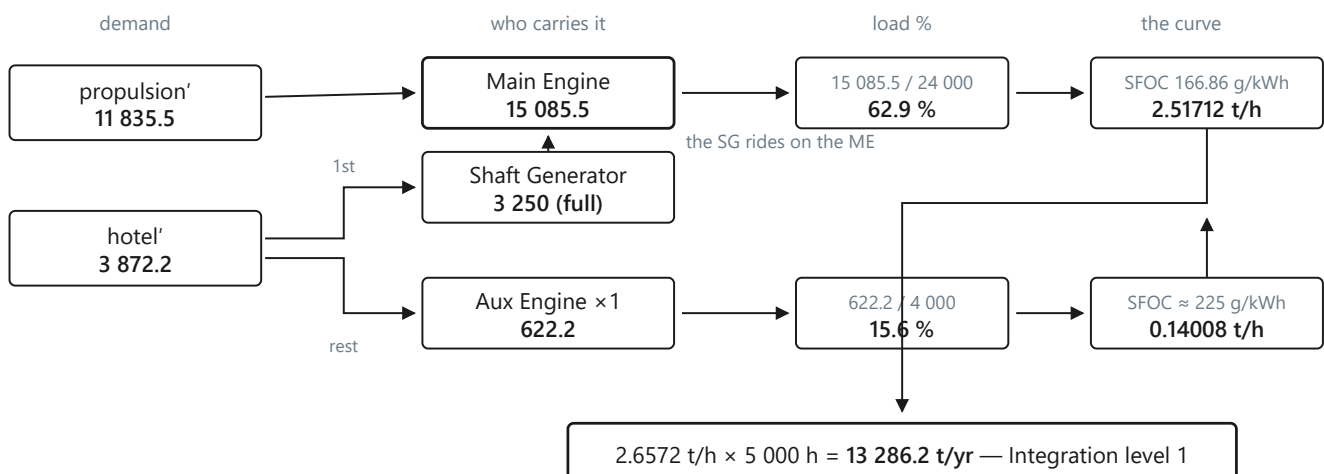
Three pictures, three claims. All numbers are scenario **01** (propulsion 11 463 kW · hotel 3 800 kW · battery 1 260 kW · Transit 5 000 h · MDO at \$780/t), so every figure here can be checked against [01-excel-baseline.pdf](#) line by line.

1 — The battery does not reduce the swing. It decides who carries it.



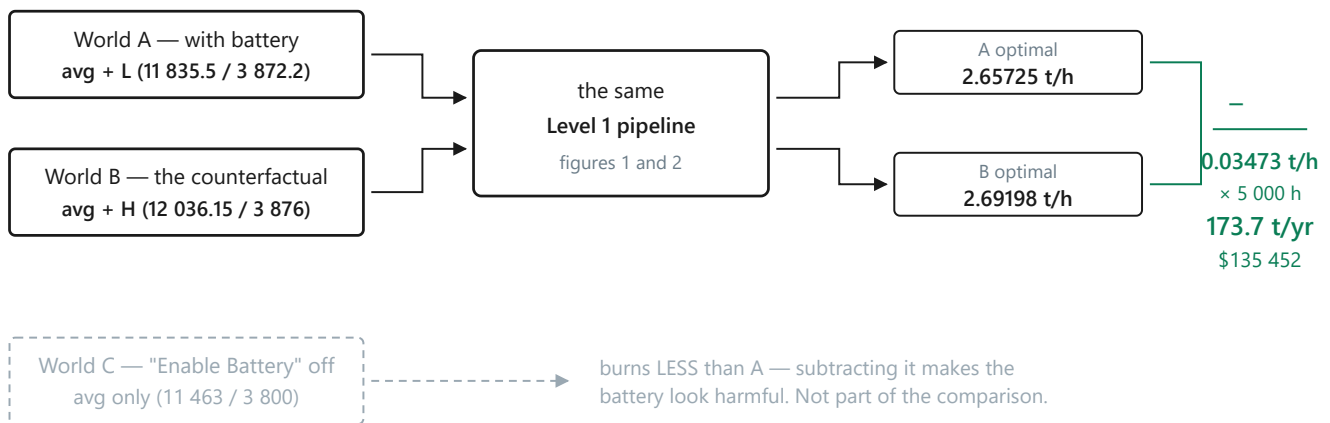
$\Sigma J + \Sigma L = \Sigma H$, always. A bigger battery moves the split; it never shrinks the 649.15. Everything the battery covers (green) stops at the tile — a covered reserve is readiness, not load, so it is never subtracted from anything. Everything it misses is added to the demand instead. This is why scenario 02, with a 300 kW battery, shows exactly the same ΣH but PS 105 / SR 544.15.

2 — Demand becomes fuel through the load split, then the curve



Two rules do all the work here. The shaft generator is filled before any auxiliary starts, because the main engine is already turning; and the SG's output is a load *on* that main engine, which is why the ME figure exceeds propulsion. The curve does the rest: the same 622 kW spread over three auxiliaries instead of one puts each at 5.2 % where the engine burns ~232 g/kWh instead of ~225 — and that difference, multiplied by 5 000 hours, is the Level 1 saving.

3 — The Battery Benefit is the same pipeline, run twice



Both figures compared are optima, never baselines — a pinned baseline is deliberately ignored in the counterfactual, or the comparison would stop being like-for-like. **World B is scenario 03**, written down as its own file: its inputs are 01's raw loads plus the full swing, which is why 03's Level 1 (13 459.9 t/yr) minus 01's (13 286.2) reproduces the badge exactly. To check a Benefit by hand, load 03 — do not untick the battery.

A number you can carry in your head

Figure 1 hands the engines **204.4 kW less** to carry; figure 2 says every kilowatt costs roughly its SFOC in fuel. So:

$$\text{Benefit} \approx \text{Peak Shaving} \times \text{SFOC} \times \text{hours}$$
$$204.4 \text{ kW} \times 167 \text{ g/kWh} \times 5\,000 \text{ h} \div 10^6 = \mathbf{170.7 \text{ t/yr}}$$

against the true 173.7

The remaining ~3 tonnes come from the engines settling at slightly better points on the curve. The approximation is close enough to sanity-check any scenario in seconds — and it shows why **CoverageFactor** is the most consequential setting in `appsettings.json`: it alone decides how much of the swing ever becomes Peak Shaving.